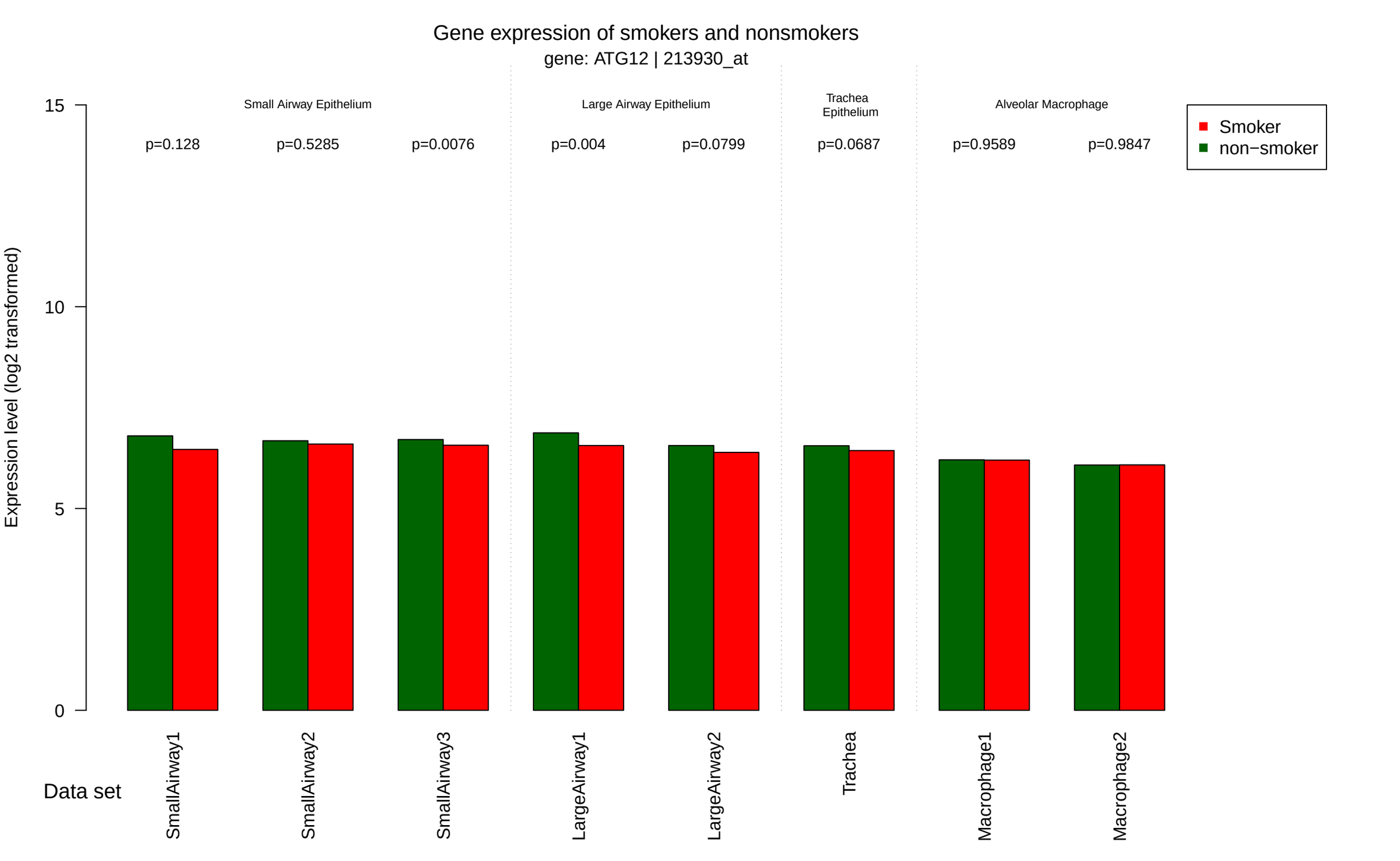
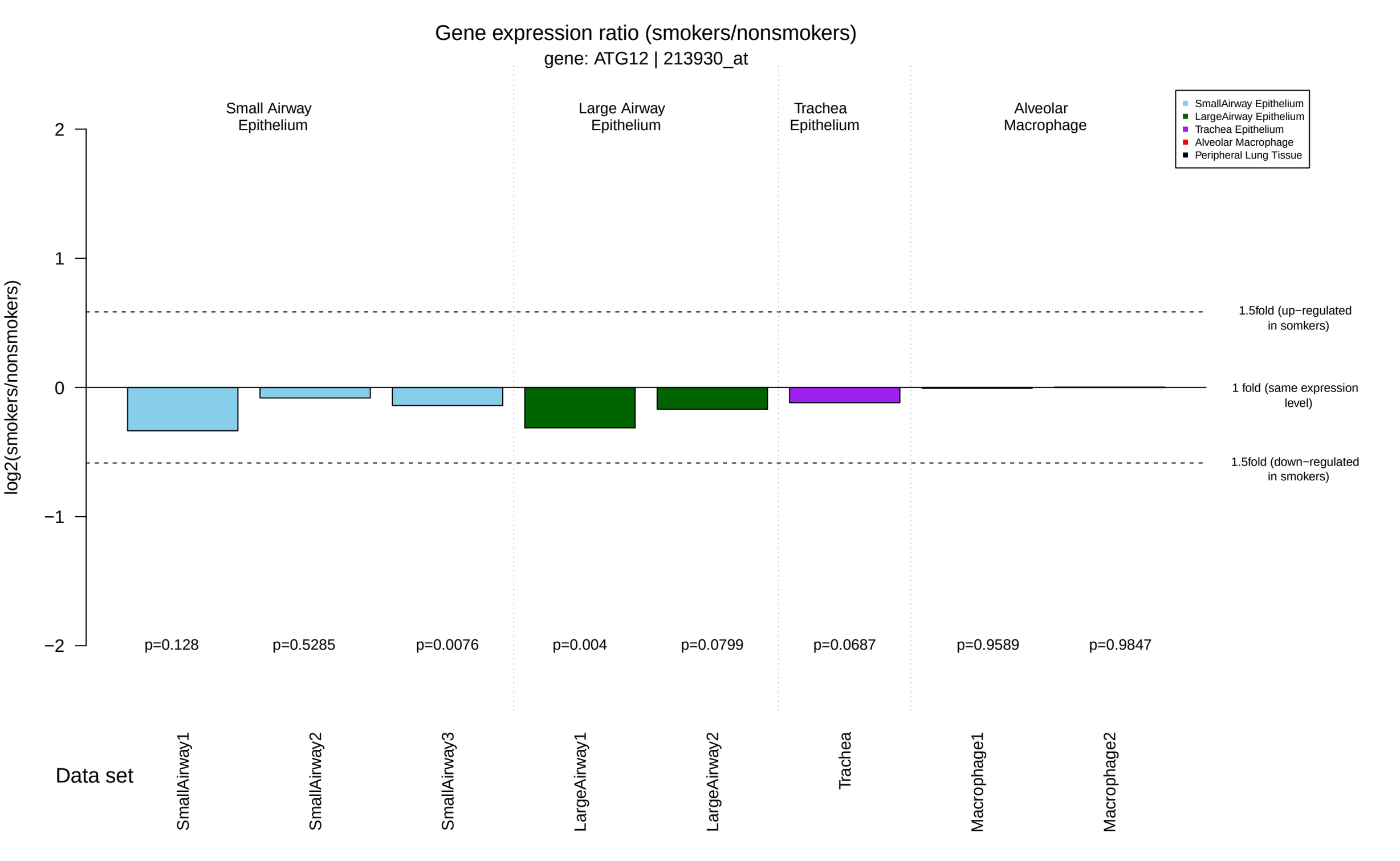
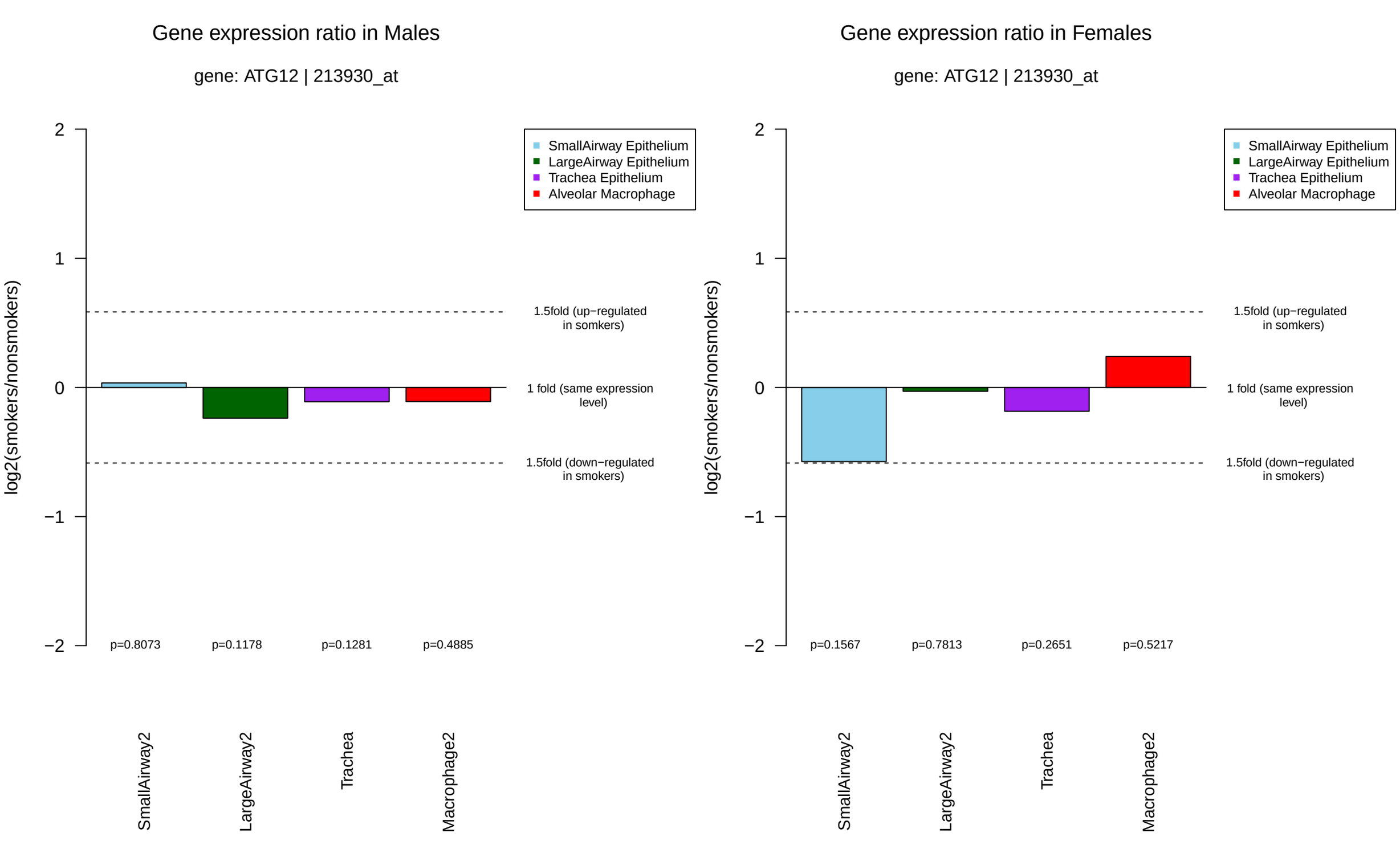
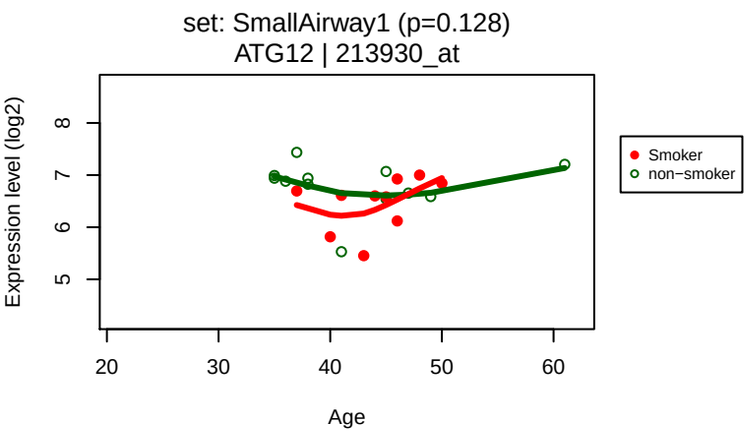
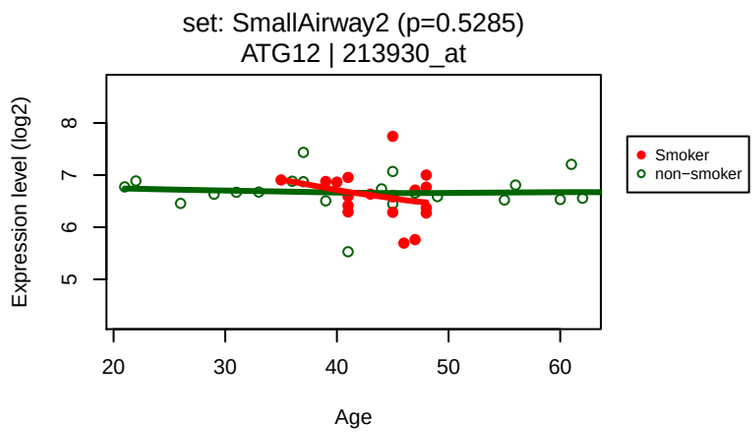
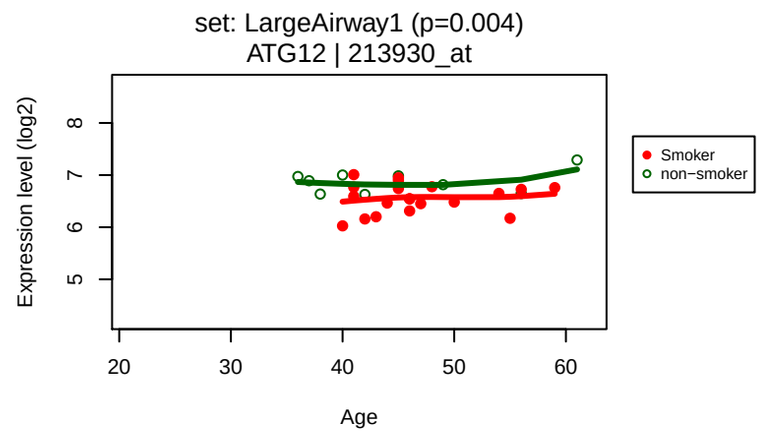

Gene expression of smokers and nonsmokers
gene: ATG12 | 213930_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.128
p=0.5285
p=0.0076
p=0.004
p=0.0799
p=0.0687
p=0.9589
p=0.9847
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

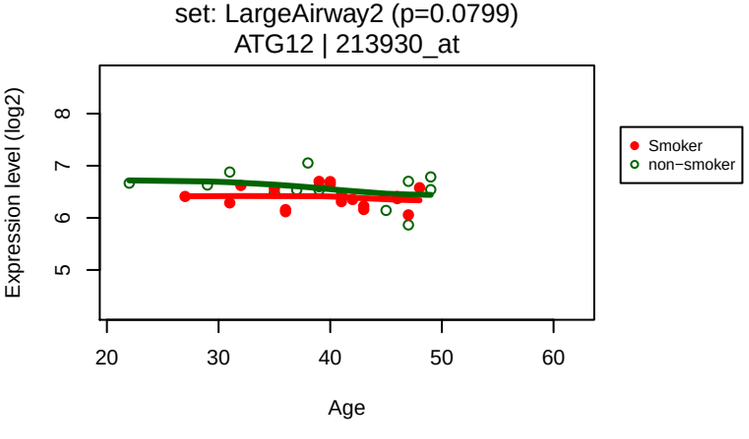

Gene expression ratio (smokers/nonsmokers)
gene: ATG12 | 213930_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.128
p=0.5285
p=0.0076
p=0.004
p=0.0799
p=0.0687
p=0.9589
p=0.9847
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

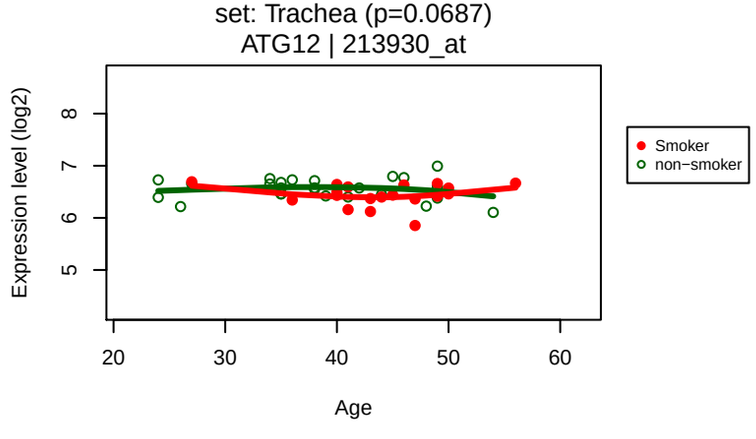

Gene expression ratio in Males
gene: ATG12 | 213930_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8073
p=0.1178
p=0.1281
p=0.4885
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATG12 | 213930_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1567
p=0.7813
p=0.2651
p=0.5217
SmallAirway2
LargeAirway2
Trachea
Macrophage2

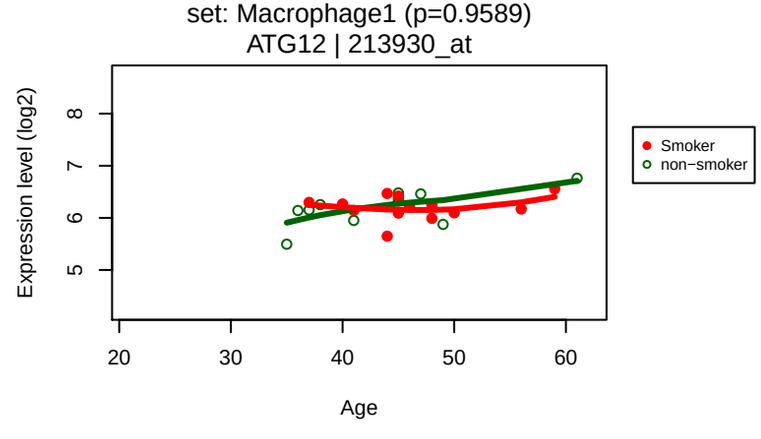

set: SmallAirway1 (p=0.128)
ATG12 | 213930_at
Expression level (log2)
Age
Smoker
non−smoker

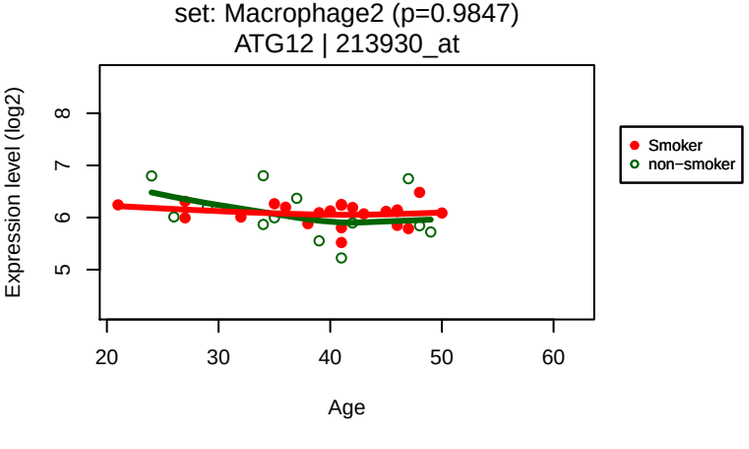

set: SmallAirway2 (p=0.5285)
ATG12 | 213930_at
Expression level (log2)
Age
Smoker
non−smoker

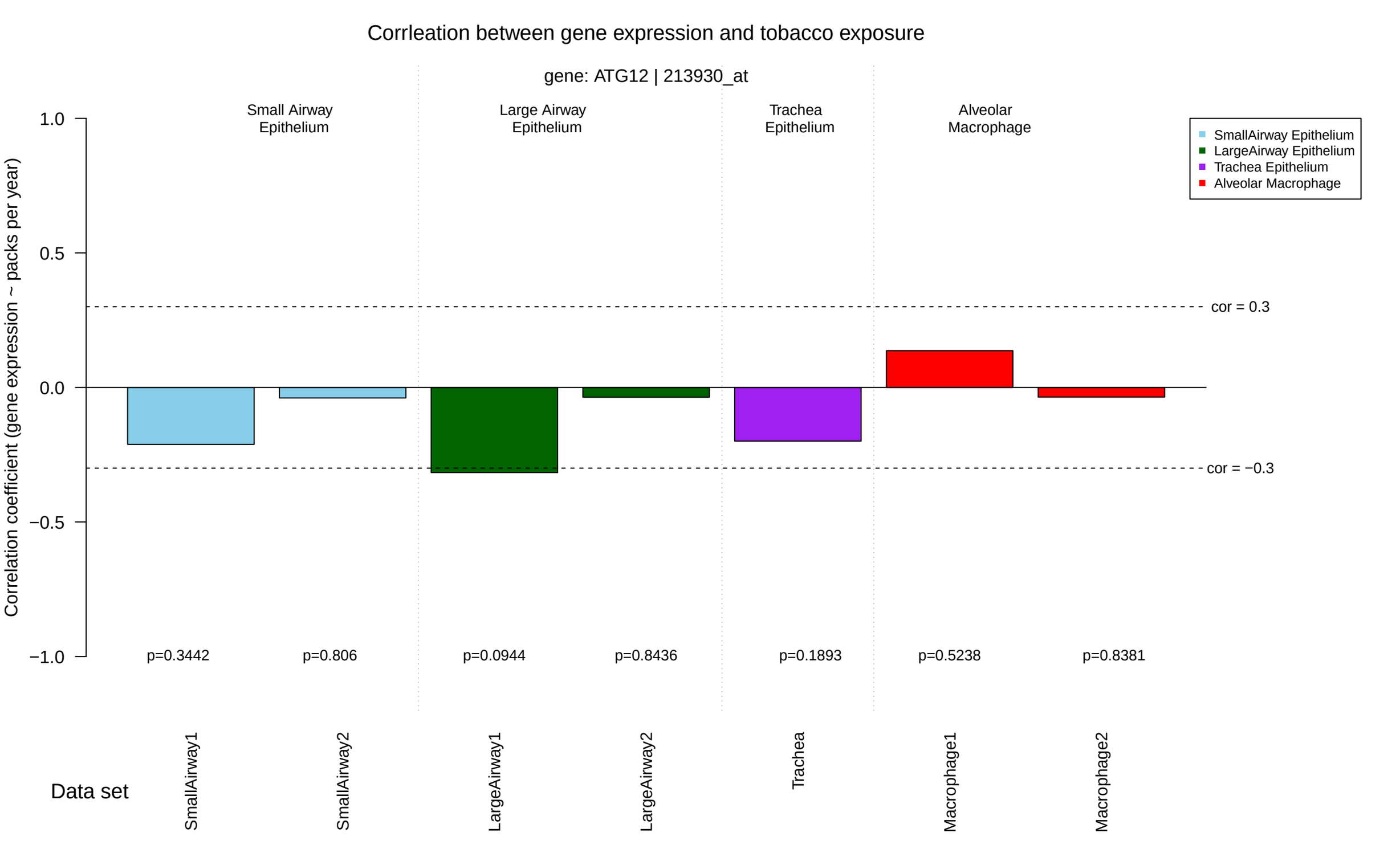

set: LargeAirway1 (p=0.004)
ATG12 | 213930_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0799)
ATG12 | 213930_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0687)
ATG12 | 213930_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9589)
ATG12 | 213930_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9847)
ATG12 | 213930_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATG12 | 213930_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3442
p=0.806
p=0.0944
p=0.8436
p=0.1893
p=0.5238
p=0.8381
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2